

FATIH KURT

AI & Automation Engineer

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SUMMARY

AI Engineer building production LLM systems: RAG pipelines, agent workflows, and process automation. Currently shipping internal AI tooling at Siemens, with a software and data engineering foundation from BMW, Continental, and Magna.

EXPERIENCE

AI & Automation Engineer, Siemens

AI tooling for an engineering department

02/2026 – present

Nuremberg

- Built and deployed a **RAG system** over the department's technical standards corpus (**500 documents**), covering ingestion, chunking, retrieval from a **Chroma** vector database, and answer generation through the internal Siemens LLM API. Tested chunking and retrieval settings against real engineer queries before rollout. Now used by **40 engineers** in place of manual PDF search.
- Built an **email classification pipeline** using LLM prompts with HTML-to-text preprocessing, deployed through **Power Automate**. Proposed the project, quantified the time saved, and presented the case to IT to get it rolled out.
- Wrote a **meeting summarizer** in Python with transcript chunking for **60+ minute** recordings, returning structured action items into the team workflow.
- Automated recurring department processes in Power Automate, including a **daily** Planner-to-Excel backup.

Software Engineer, Data & Analytics, BMW

Intern → working student, BMW & MINI digital customer channels

08/2024 – 02/2026

Munich

- Built and maintained **KPI dashboards** in Tableau, Power BI, and Streamlit for the BMW and MINI digital customer channels, used by around **150 people** across several agile product teams.
- Automated data pipelines on **AWS** and **Snowflake** pulling from **30+ source systems** into roughly **200 tables**; ran SQL and Python analyses to surface trends behind product decisions.
- Wrote a **monitoring script for test results** with email alerting, so failures show up immediately rather than in manual checks.
- Built **Streamlit** apps for data exploration and refactored existing code for performance and maintainability.

Embedded Software Engineer, Head-Up Displays, Continental

Working student, embedded systems

02/2024 – 08/2024

Ingolstadt

- Wrote and optimized **C/C++ modules** for the **MX8 board** in Head-Up Display systems.
- Built custom Linux distributions with **Yocto** for the MX8 board and set up **SSH remote access** for the team's development boards.

Python Test Automation Engineer, Magna

Working student, System Validation Driving Functions & E-Mobility

06/2023 – 02/2024

Ingolstadt

- Wrote executable **Python test cases** from test designs and maintained specifications in **DOORS** for safety-relevant vehicle functions.
- Tested **Audi brake and steering components** with EXAM and ControlDesk.

PROJECTS

B2B AI Concierge for Boutique Hotels | n8n · OpenRouter · Telegram Bot API · Python

09/2025 – present

- Multilingual** AI concierge for hotels featuring **multi-tier LLM routing** to optimize inference costs. With the technical MVP fully deployed, current efforts are focused on B2B marketing and onboarding first pilot clients.

Bachelor Thesis: RL for Adaptive Traffic Signal Control | Python · SUMO/TraCI · PPO · MLflow · Pygame 10/2025 – 03/2026

- Framework where **PPO agents** and human players control traffic signals side by side in SUMO, with a live Pygame visualization. Tracked training runs and hyperparameter sweeps in **MLflow** and compared agent performance against human performance across scenarios. Grade **1.3**.

- Real-time **time-to-collision ROS 2 nodes** with perception integration, tested in simulation **and on the actual vehicle**. Also worked on a Pure-Pursuit controller and the team’s Docker setup for reproducible installs.
- Bosch Mobility Challenge & Teknofest Racing | OpenCV · YOLO

2022 – 2024
- Road detection and real-time object recognition for autonomous vehicle perception, in multidisciplinary teams.

SKILLS

LLM Engineering: RAG (chunking, embeddings, retrieval), LangChain, CrewAI multi-agent systems, Chroma vector database, OpenRouter model routing, Ollama (self-hosted LLMs), prompt engineering, LLM evaluation

MLOps & Deployment: MLflow (experiment tracking), Docker, REST API deployment (FastAPI), Git/GitHub, automated monitoring & alerting

Automation: n8n, Microsoft Power Automate, Telegram Bot API, workflow design

Programming: Python (FastAPI, REST APIs), C/C++ , SQL

Data & Cloud: AWS, Snowflake, Power BI, Streamlit, Tableau

Systems & Robotics: Linux, Yocto, ROS 2, Gazebo, SUMO

ML / Computer Vision: Reinforcement Learning (PPO), YOLO, OpenCV, TensorFlow

EDUCATION

B.Eng. Autonomous Vehicle Engineering, Technische Hochschule Ingolstadt
Thesis (Grade: 1.3): Reinforcement Learning for adaptive traffic signal control

10/2022 – 07/2026
Ingolstadt

German Language Courses, Universität des Saarlandes

10/2021 – 08/2022
Saarbrücken

LANGUAGES & CERTIFICATIONS

Languages: Turkish (native), German (C1), English (C1)

Certifications: Machine Learning Specialization (Coursera): supervised & unsupervised learning, recommender systems, RL; MATLAB Onramp (MathWorks)